

JOÃO GUILHERME SIMIONATO

Applied AI Engineer & Full Stack Developer · Computer Engineer

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PROFESSIONAL SUMMARY

Computer Engineer specializing in backend AI engineering, LLM productization, and high-throughput audio processing pipelines. Experienced in optimizing deep learning inference (Faster-Whisper), building FastAPI-based AI microservices, architecting speaker diarization, and deploying scalable RAG workflows. Combines strong Python backend expertise with Full Stack capabilities (React, TypeScript) to bridge machine learning models with production-grade client interfaces.

PROFESSIONAL EXPERIENCE

Full Stack AI Developer | EZ Voice

Apr 2026 – Present

- Inference & Batch Optimization:** Re-engineered speech-to-text pipelines using dynamic batching on Faster-Whisper, achieving a 1.8x speedup (~45% latency reduction) with 100% benchmark processing reliability across test datasets (80/80 audio files).
- Audio Processing (Python/NumPy):** Engineered high-performance audio preprocessing pipelines for long-form call recordings, implementing normalization, chunking, and memory-optimized array operations.
- Containerization & Deployment:** Used Docker to containerize selected AI applications, standardizing runtime environments and simplifying deployment.
- Speaker Diarization:** Modeled and integrated backend workflows to separate customer and agent voices, improving call analytics and conversational transcript accuracy.
- LLMs & RAG Architectures:** Designed intelligent retrieval workflows and implemented model reliability metrics (log probabilities, error rates) to calculate system confidence scores, exposing AI workflows through FastAPI services.
- Front-End Integration (React/TS):** Connected AI microservices to React/TypeScript interfaces, delivering seamless end-to-end integration and responsive UI feedback.

Full Stack Developer | EZ Voice

Aug 2024 – Apr 2026

- Frontend Development:** Developed modern, responsive web interfaces using JavaScript, TypeScript, and React, emphasizing clean code, modular architecture, and accessibility standards.
- REST APIs & State Management:** Built and integrated FastAPI-based REST APIs and optimized application state management for seamless asynchronous user workflows.
- Backend & PHP Integration:** Collaborated on backend API routing and data structuring using PHP, streamlining business logic and client-server communication.

TECHNICAL SKILLS

AI & Audio ML: Faster-Whisper, Batch Inference Optimization, Speaker Diarization, NumPy, LLM Integration, RAG Architectures, LangChain, Hugging Face, OpenAI API

Languages & Backend: Python, TypeScript, JavaScript (ES6+), PHP, SQL, REST APIs, FastAPI, PostgreSQL, Redis, Microservices Architecture

Frontend & Web: React, State Management, HTML5, CSS3, Modern UI/UX, Responsive Design

Tools & Infrastructure: Git, GitHub, Docker, Linux, ESP32 Microcontrollers, IoT Sensors (BME280), MLOps Foundations

EDUCATION

Postgraduate Degree in Generative AI & LLMs | PUC Minas

2026 – 2027

- Agent Engineering, Multimodal LLMs, Deep Learning Architectures, and MLOps.

Bachelor of Science in Computer Engineering | UNAERP – University of Ribeirão Preto

2021 – 2025

CERTIFICATIONS

Python for AI & Automation | Asimov Academy

Jun 2025

- Practical training in Python for AI and application development.
- Experience with LangChain, OpenAI API, and Hugging Face.

JavaScript & TypeScript — Basic to Advanced

2024

- Advanced JS (ES6+) and TypeScript covering Functional Programming, OOP, and modern design patterns.

PROJECTS

IoT Weather Station | UNAERP – University of Ribeirão Preto

Aug 2024 – Dec 2024

- Developed a connected IoT weather station for measuring temperature, humidity, and wind speed.
- Implemented using ESP32 microcontroller, BME280 sensors, and Python/JavaScript integration.
- Presented and published at the UNAERP Scientific Initiation and Research Congress (CONIC).